9.2 Special Right Triangles

Name _____ Period ____ Date _____

Geometry 9.2 (30-minute plan) · Student sheet

I can…

- find the side lengths of a 45°-45°-90° triangle from one side.
- find the side lengths of a 30°-60°-90° triangle from one side.
- use special right triangles to solve real-life problems.

Vocabulary · fill in the missing word

45°-45°-90° triangle — an _____ right triangle; its two legs are congruent

30°-60°-90° triangle — a right triangle with _____ of 30°, 60°, and 90°; half of an equilateral triangle

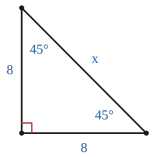
Shorter leg — in a 30°-60°-90° _____, the leg across from the 30° angle

Part 1 · Notes and examples (fill in each note, then work the example)

45°-45°-90° Triangle Theorem: the hypotenuse is _____ times as long as each leg: hypotenuse = leg · √2.

Example 1

Find the length of the hypotenuse.

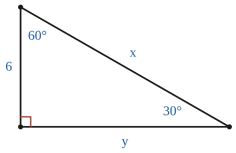


From the hypotenuse: to find a leg, _____ the hypotenuse by √2, then rationalize.

30°-60°-90° Triangle Theorem: the hypotenuse is _____ times the shorter leg, and the longer leg is _____ times the shorter leg.

Example 2

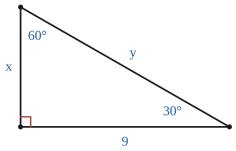
The shorter leg of a 30°-60°-90° triangle is 6. Find the hypotenuse and the longer leg.



Shorter leg first: in a 30°-60°-90° triangle, always find the _____ leg first.

Example 3

The longer leg of a 30°-60°-90° triangle is 9. Find the shorter leg and the hypotenuse.



Part 2 · Partner practice

Partner practice

The hypotenuse of a 30°-60°-90° triangle is 14. Find the shorter leg and the longer leg.